Chapter 1 - What Is a DBA?

1. Database Administrator - DBA

Definition: An information technician responsible for the ongoing operational functionality and efficiency of an organization's databases and the applications that access them.

A DBA does more than create databases. The DBA ensures that databases keep operating efficiently, applications can use them correctly, and data remains available, secure, and recoverable.

Key Points: DBA = operational functionality + efficiency.

Exam Focus: "A DBA is responsible only for database security." **False.** The role includes continued database and application operation and efficiency.

2. Database vs. DBMS

Database: An organized store of data accessible through named data elements. In simple terms, it is the organized, stored data.

Database Management System (DBMS): Software used to create, store, retrieve, update, share, and manage data in databases. Oracle, Microsoft SQL Server, and IBM DB2 are DBMS products, rather than the database itself.

A DBMS also supports data integrity, security, access control, optimization, rollback, restart, and recovery.

Key Points: Database = data. DBMS = software managing the data.

Exam Focus: "A DBMS is software that enables users or programmers to share data." **True.**

3. Reactive DBA vs. Proactive DBA

Database administration should be a management discipline: planning the work and then carrying it out according to that plan.

- **Reactive DBA:** Acts after a problem occurs. Example: performance collapses, and only then does the DBA investigate. This approach resembles firefighting.
- **Proactive DBA:** Plans ahead and uses procedures and practices to prevent problems. Planning leads to prevention and fewer problems.

Key Points: Reactive = after the problem, firefighting, fixing problems. Proactive = before the problem, planning, preventing problems.

Exam Focus: "A proactive DBA waits until database problems occur before taking action." **False.** That describes a reactive DBA.

4. Application Development Life Cycle - ADLC

The DBA should participate throughout application development:

Requirements > Analysis > Design > Development > Testing > Production > Maintenance.

Early involvement clarifies data requirements. The DBA contributes to conceptual and logical design, translates the logical design into a physical database, helps with testing, and supports movement from test

to production. After deployment, the DBA continues to handle performance, availability, backup and recovery, and security.

Exam Focus: "DBAs should participate only after the application reaches production." **False.** "DBAs should participate from requirements gathering through deployment." **True.**

5. Data, Database, and System Administration

Data Administration (DA) emphasizes the business aspects of data: requirements, policies, ownership, conceptual models, logical models, and metadata. Remember: **DA = business + data.**

Database Administration (DBA) emphasizes technical database management: physical design, performance, availability, backup, recovery, and security. Remember: **DBA = databases + technology.**

System Administration (SA) emphasizes DBMS installation and system infrastructure: installation, upgrades, system software, operating-system configuration, and infrastructure. Remember: **SA = system / DBMS infrastructure.**

Exam Focus: "Data administration manages business aspects of data, while database administration focuses on technology." **True.** This is a recurring multiple-choice distinction. Organizational responsibilities can overlap.

6. DBA Tasks

Principal DBA tasks include:

- Database design.
- Performance monitoring and tuning.
- Ensuring availability.
- Database security and authorization.
- Governance and regulatory compliance.
- Backup and recovery.
- Ensuring data integrity.
- DBMS release migration.

Key Points: Governance and regulatory compliance are DBA responsibilities.

7. Database Design

The DBA needs to understand conceptual models, logical models, and physical databases. The crucial transition is **logical data model > physical database design**: implementing a logical design in a real DBMS.

The DBA also needs relational theory and entity-relationship diagrams (ERDs). Design matters, but it usually occupies less DBA time than ongoing administration and tuning.

Exam Focus: "The DBA transforms the physical implementation into the logical model." **False** for the design direction being tested. Remember **logical > physical**.

8. Performance Monitoring and Tuning

Memorize the five factors: **workload, throughput, resources, optimization, and contention.**

Workload: The work requested from the DBMS at a particular time. It includes online transactions, batch jobs, ad hoc queries, data warehousing or analytics, and commands. **Workload = demand.**

Throughput: The overall hardware and software capability to process data. It is affected by CPU speed, I/O speed, parallelism, and operating-system efficiency. **Throughput = processing capability.**

Resources: Available system capabilities, including disk space, the database kernel, cache, and other hardware and software resources.

Optimization: Selecting an efficient way to execute a database request or query. The objective is an efficient access path.

Contention: Conflicting attempts by components to use the same resource. Example: two operations attempt to modify the same data. As contention increases, throughput can decrease.

Key Points: Database performance involves optimizing resource use to increase throughput and minimize contention.

Exam Focus: Overall processing capability = **throughput**. Amount or type of requested work = **workload**. Conflicting use of the same resource = **contention**.

9. Availability

Availability means that the DBMS and databases can be used when needed. Keeping the DBMS running is the starting point. The DBA also minimizes downtime, performs maintenance with minimal disruption, uses monitoring and alerts, plans for failover, and preserves availability during administrative tasks.

Key Points: Availability and performance are related but different. A database can be available yet slow.

Exam Focus: "Availability and performance mean exactly the same thing." **False.**

10. Database Security and Authorization

Only authorized users should access or modify data. The DBA manages privileges using mechanisms such as **GRANT** and **REVOKE**.

Security can govern creating or altering database objects, reading or modifying data, executing stored procedures, running utilities, and changing DBMS parameters.

Key Points: Security asks **who can do what**, rather than only who can log in.

Exam Focus: "Database security only controls who can read data." **False.** It controls many types of actions.

11. Governance and Regulatory Compliance

The DBA helps implement the technological controls needed to comply with laws and organizational policies. The DBA does not independently define the law or compliance standard.

Examples include security controls, auditing, backup requirements, change management, data retention, data masking, and encryption.

Key Points: Business and legal functions establish applicable compliance requirements; the DBA helps implement technical controls.

Exam Focus: "The DBA alone decides which regulations the company must follow." **False.**

12. Backup and Recovery

The DBA must be prepared to restore data after hardware failure, software error, human error, or disaster. Distinguish three recovery objectives:

- **Recover to current:** Return the database to its state at the time of failure.
- **Point-in-time recovery:** Return the database to a specific past moment, removing the effects of later transactions. Example: a problem happens at 4:10, but the desired recovery point is 4:00.
- **Transaction recovery:** Remove the effects of particular transactions rather than moving the entire database back to an earlier time. It is also called application recovery.

Key Points: Current = failure-time state. Point-in-time = earlier timestamp. Transaction = selected transactions.

Exam Focus: Point-in-time recovery restores a database to a particular past moment to undo unwanted changes.

13. Data Integrity

Integrity means storing correct data in the correct way without corruption. Memorize **physical, semantic, and internal integrity**.

Physical integrity: Data conforms to its defined type and constraints. An INTEGER column must contain valid integer values, rather than arbitrary text. Mechanisms include data types, domains, and constraints.

- **Referential constraint:** Preserves valid relationships between tables.
- **Unique constraint:** Prevents duplicate values or combinations of values that must be unique.
- **Check constraint:** Restricts values according to a specified rule.

Semantic integrity: Data is meaningful and correct in terms of business meaning and quality. A customer's address may be valid text but still be the wrong address. The type can be correct while semantic quality is poor. Remember **data quality**.

Internal integrity: The DBMS maintains sound internal structures such as indexes, pointers, and other internal structures.

Exam Focus: Physical = types / constraints. Semantic = meaning / data quality. Internal = DBMS structures.

14. DBMS Release Migration

The DBA manages movement to a newer DBMS version or release. The objective is not to upgrade immediately every time. Balance new features against risk, downtime, compatibility, and cost.

A new release can improve performance, add features, and fix bugs. It can also cause downtime, break compatibility, change access paths, or require application changes.

Exam Focus: An upgrade strategy balances new-feature benefits against disruption and incompatibility risks.

15. Types of DBAs

System DBA: Focuses on technical and system issues: DBMS installation, upgrades, system parameters, storage, and OS or network interaction. **Focus: the overall DBMS.**

Database architect: Designs and implements new databases. **Focus: new database design**, rather than primarily maintaining existing databases.

Database analyst: Has no universal fixed definition. Depending on the organization, this may mean a junior DBA, an architect-like role, or another DBA title.

Data modeler: Identifies data requirements and develops conceptual and logical data models.

Application DBA: Supports a specific application or application group. Often strong in SQL, application support, and tuning. System DBA = overall environment; application DBA = particular applications.

Task-oriented DBA: Specializes in one DBA task, such as backup and recovery. More common in large organizations.

Performance analyst: A task-oriented DBA specializing in performance, requiring strong knowledge of SQL performance and DBMS parameters.

Data warehouse administrator: Specializes in the warehouse environment and understands the differences between online transaction processing (OLTP) and warehouse or analytical systems.

Key Points: Match each title to its focus: system, new design, organization-dependent analysis, models, applications, one task, performance, or warehouse.

16. Staffing Considerations

There is no fixed correct number of DBAs. Staffing depends on the number of databases, users, and applications; availability and performance requirements; the impact of downtime; DBA experience; developer experience; and available tools and automation.

Key Points: Determining DBA staffing is not an exact science.

17. Multiplatform DBA Issues

A multiplatform organization can use multiple DBMS products on multiple platforms, such as Oracle, DB2, and SQL Server on different systems. Each has different features, tools, and behavior, increasing administrative complexity.

Exam Focus: "Supporting multiple DBMS platforms generally simplifies database administration." **False.** It increases complexity.

18. Production vs. Test

Test environment: Used for development, testing, and maintenance activities.

Production environment: Runs real operational applications and business activity.

Separate the environments to protect integrity and performance. A program being tested must not accidentally modify production data. Additional stages can include unit testing, integration testing, and quality assurance (QA), followed by production.

Exam Focus: "Test and production environments should be merged to simplify development." **False.** This distinction is tested in the practice material.

19. Procedural DBA

Some application logic resides inside the database as stored procedures, triggers, and user-defined functions (UDFs).

Stored procedure: A program stored and executed in the database environment. It can execute several operations on the server before returning results, reducing repeated client/server exchanges and potentially reducing network traffic.

Trigger: An action executed automatically in response to an event, such as INSERT, UPDATE, or DELETE. Remember **trigger = event-driven**.

UDF: A function created by a user and used as a function within the DBMS.

Procedural DBA: Manages stored procedures, triggers, and UDFs, overseeing planning, implementation, testing, performance, and reuse.

Exam Focus: "Stored procedures can reduce network traffic by executing multiple server-side operations before returning data." **True.** "Triggers are manual procedures that the DBA must run." **False.**

20. eDBA

An eDBA is a DBA capable of managing Web-based and Internet-enabled database applications. The role requires understanding 24/7 availability, Web connectivity, Java and XML, legacy integration, unpredictable workloads, and Internet performance.

Key Points: eDBA = Internet / Web. It does not mean an email DBA or simply an electronic DBA.

21. Cloud Computing

Cloud computing provides on-demand computing resources without requiring the organization to buy all of the underlying hardware. It commonly uses a pay-as-you-go model: payment based on usage.

DBA concerns include data availability, performance, synchronization, remote users, security, and large quantities of data.

Exam Focus: "Cloud computing has no impact on DBA responsibilities." **False.**

22. NoSQL and Big Data

NoSQL is often associated with nonrelational, distributed, flexible, and scalable systems. Many NoSQL systems use less rigid schemas, support replication, emphasize scalability, and are associated with Big Data.

NoSQL does not necessarily mean absolutely no SQL. Some products provide SQL-like querying, which is why the term is sometimes interpreted as **Not Only SQL**.

Big Data creates a need to store and access large amounts of data with suitable scalability and cost.

Exam Focus: "NoSQL databases require strict schemas for all data." **False.** NoSQL systems still require administration.

23. DBA Certification

Certification demonstrates knowledge of DBA tasks, but certification does not automatically make someone a better DBA. Practical experience matters. Certification may improve employability, and some organizations prefer or require certified professionals.

Exam Focus: "Certification guarantees that a person is an excellent DBA." **False.** "Certification may improve employability." **True.**

24. DBA Communication Skills

DBAs need exceptional communication skills. They interact with programmers, technicians, end users, customers, and executives. Their involvement throughout systems development requires understanding technical problems and business needs.

Exam Focus: "DBAs only work on technical database matters and never interact with business users or executives." **False.**

25. Database Administration Is a Nonstop Job

Critical databases and applications may operate 24/7. A DBA can therefore be on call and work outside normal hours or on weekends, especially during failures, performance problems, and maintenance.

Exam Focus: Why do DBAs sometimes work long hours and remain on call? **Because database administration is nonstop and business operations continue around the clock.**

26. Privacy Policies and Data

Privacy policies and applicable requirements affect data collection, storage, access, transfer, retention, and disposal. Privacy must be considered from design through operation to the end of the data or application's life.

A policy or regulation may require a particular retention period or disposal method. The DBA's role does not begin only after deployment.

Key Points: Privacy policy and regulatory compliance are related but are not identical concepts.

Exam Focus: "DBAs help ensure that data handling complies with privacy regulations during design and operation." **True.**

27. e-Vailability

e-Vailability means the very high availability expected of Web-enabled and Internet businesses. Users may connect from anywhere at any time, so the target is 24/7 availability with minimal downtime.

Key Points: eDBA = the person / role. e-Vailability = the high availability requirement for the Web.

Exam Focus: A primary concern for databases serving Internet users is **high availability and minimal downtime**, or e-vailability.

28. Multiplatform DBA Staffing

For a multi-DBMS organization, employ platform-expert DBAs for the major DBMS products and share specialists for less-used platforms. Do not assume one DBA knows every DBMS perfectly.

Exam Focus: A recommended staffing approach is **platform-expert DBAs for each major DBMS, with shared specialists for less-used platforms**.

Chapter 2 - Creating the Database Environment

1. Defining the Organization's DBMS Strategy

A DBMS strategy is the organization's overall plan for selecting, using, and supporting DBMS technology. Selection should be coordinated rather than random or independently driven by each department.

Key Points: The DBA group should be empowered to make DBMS decisions for the organization. Unplanned product diversity makes support harder, increases complexity and cost, creates compatibility problems, and adds DBA work.

Exam Focus: "Each business unit should independently choose its own DBMS." **False.** "The DBA group should be empowered to make DBMS decisions." **True.**

2. Choosing a DBMS

Minimize the number of different DBMS products. Select a default DBMS for each platform and make exceptions only for a compelling business case. Do not choose a DBMS solely because of one attractive new feature.

Tier-1 Vendors

The principal Tier-1 vendors/products are IBM DB2, Oracle, and Microsoft SQL Server. These are the largest, widely supported choices in this context; choosing an established vendor generally reduces vendor-related risk.

Exam Focus: "Tier-1 vendors are the largest and most supported in the market." **True.**

DBMS Selection Factors

- **Operating-system support:** Is the DBMS supported on the required platform and OS?
- **Type of organization:** Is the organization conservative, aggressive, or comfortable with risk? Does it rely on mainframe, UNIX, or Windows environments?
- **Benchmarks:** TPC benchmarks help evaluate performance. **TPC-C** represents transaction processing; **TPC-H** covers decision support and ad hoc queries; **TPC-E** is a newer OLTP workload. TPC means Transaction Processing Performance Council.
- **Scalability:** Can the system grow with data volume and user numbers?
- **Supporting tools:** Are backup, recovery, monitoring, administration, and query tools available?
- **Technicians:** Are skilled DBAs, programmers, and support staff available?
- **Total cost of ownership (TCO):** Include licenses, supporting software, DBA and programmer costs, and computing resources. Hardware and operational costs also matter.
- **Release schedule:** Frequent releases can be beneficial or disruptive depending on the organization.
- **Reference customers:** Ask current customers about support, bugs, performance, and stability.

Exam Focus: "Choose a DBMS solely for its newest feature." **False.** "Consolidating around a default DBMS simplifies management." **True.** A vendor logo's color is not a meaningful TCO component.

3. DBMS Architectures

Memorize the four architecture levels in order:

Enterprise DBMS: Supports very large databases, many concurrent users, high performance, and advanced features. Associate it with large-scale, mission-critical processing.

Departmental DBMS: Also called a workgroup DBMS. Supports small to medium workgroups at a smaller scale than an enterprise DBMS.

Personal DBMS: Designed for a single user and very small projects. It is unsuitable for multiuser applications.

Mobile DBMS: Designed for remote users who are not always connected. Supports local data access and modification, followed by synchronization with a central database.

Exam Focus: Enterprise = very large databases + many users. Personal = one user. Mobile = remote use + synchronization. "A personal DBMS is suitable for multiuser enterprise applications." **False.**

4. DBMS Clustering

Clustering uses multiple systems or processors working together as one system. Its goals include availability, scalability, failover, and load balancing.

Shared-nothing clustering: Each node has private resources, including memory and disk, and nodes communicate over a network. Its main advantage is scalability; nodes are comparatively independent.

Shared-disk clustering: Nodes have private memory but share disks. It is associated with large enterprise processing, mainframe-like environments, and workloads that can be difficult to partition.

Key Points: Shared-nothing = private resources + network communication + scalability. Shared-disk = private memory + shared disks. Suitability depends on the workload.

Exam Focus: "Clustering enhances availability and scalability." **True.** "The main advantage of shared-nothing is scalability." **True.** "Shared-nothing means all systems share disks." **False.**

5. DBMS Proliferation

Proliferation means having many different DBMS products in the organization. Every product requires administration, support, knowledge, tools, and maintenance. Unnecessary variety increases complexity and cost and makes support harder.

Key Points: Establish a policy before introducing another DBMS product.

Exam Focus: "The fewer DBMS products installed, the easier database administration becomes." **True.**

6. Hardware Issues

DBMS selection must consider the hardware platform, operating system, reliability, availability, and scalability. Hardware and OS choices directly affect the success of the database environment.

Exam Focus: "Hardware and operating-system constraints must be part of DBMS selection." **True.**

7. Cloud Database Systems

Cloud database systems provide database and DBMS services over the Internet using an external provider rather than requiring all infrastructure to be maintained locally.

Advantages: Lower initial cost, less local administration, greater business focus, easier data sharing, and reduced local installation and patching effort.

Disadvantages / risks: Data resides with an outside party; direct control is reduced; security and trust must be evaluated; and misuse or fraudulent claims by customers may be possible.

Exam Focus: "Cloud database systems store and manage data on the vendor's remote servers." **True** for the model described. "Cloud computing has no impact on DBA responsibilities." **False.**

8. Installing the DBMS

Installation is not a trivial task. Begin by understanding prerequisites: supported OS version, memory requirements, compatibility with related software, and dependencies.

Read the installation guide from beginning to end. During the initial installation, work closely with the vendor or a qualified consultant.

Exam Focus: "Work with the vendor or a qualified consultant during initial installation." **True.** OS compatibility is an installation prerequisite.

8.1 Hardware Requirements

A DBMS has minimum CPU needs, supported hardware requirements, and sometimes different editions or flavors for different uses. Choose the correct DBMS flavor and match the hardware to its requirements.

Exam Focus: "Choose the correct DBMS flavor and match hardware to requirements." **True.**

8.2 Storage Requirements

Plan storage for more than user data:

- System catalog / data dictionary and system databases.
- Log files and archived logs.
- Start-up and control files.
- Work files and default databases.
- Temporary structures.
- Dump and error files.
- DBA databases.

Account for every DBMS storage requirement. Multiple storage devices can improve performance. Logs may be moved into archives and must be retained when needed for recovery. External storage, such as tape and optical media, can hold backups and off-loaded logs.

Exam Focus: "Off-loading logs to external storage is unnecessary." **False.** "Storage planning includes logs, the catalog, temporary structures, and system files." **True.**

8.3 Memory Requirements

The DBMS uses memory for the buffer pool / data cache, program cache, locks, sorting, optimization, and SQL processing. Sufficient, appropriately allocated memory reduces physical I/O and can improve performance.

Key Points: Provide more than merely the minimum memory required to start the DBMS.

Exam Focus: "Reading from memory is cheaper and faster than reading from disk." **True.** "Larger buffer pools can improve performance by reducing physical I/O." **True.**

8.4 Configuring the DBMS

Configuration means setting system parameters. These govern memory usage, logs, caching, authorization, feature enablement, and general DBMS behavior.

Do not blindly rely on default parameters. Unsuitable defaults can cause poor performance, configuration problems, and persistent operational issues.

Exam Focus: "Always rely on default system parameters after installation." **False.**

8.5 Connecting Supporting Infrastructure Software

Supporting infrastructure includes networks, transaction-processing monitors, message queues, middleware, programming languages, systems management software, operations software, Web servers, and application servers.

After installing the DBMS, ensure that it connects correctly to the surrounding environment. Each component may need its own configuration.

Exam Focus: "Each supporting software component may require separate configuration to work with the DBMS." **True.**

8.6 Installation Verification

Use sample programs, vendor verification procedures, interactive SQL, or scripts that exercise SELECT, INSERT, UPDATE, and DELETE.

Completion of the installation process alone is insufficient: verify that the DBMS actually works correctly.

Exam Focus: "Installation should be verified with testing, rather than assumed successful." **True.**

8.7 DBMS Environments

An installation plan normally includes multiple environments, such as testing, QA, integration, and production. Use separate DBMS installations / environments rather than mixing all activities together.

Multiple environments support development, minimize migration issues, and make testing, tuning, and monitoring easier.

Exam Focus: "Multiple environments support development and minimize migration issues." **True.** "Testing and production should be merged for faster development." **False.**

9. Upgrading DBMS Versions and Releases

Change is a fact of life. The DBA needs an explicit upgrade strategy.

Version: Usually a major change with more features and greater impact.

Release: Usually a smaller incremental change, although it may still be significant.

Exam Focus: "A version change is usually bigger than a release change." **True.**

Benefits and Risks

Benefits: New features; application or vendor requirements; improved performance; improved availability; better vendor support; possible cost savings; and alignment between test and production.

Risks: Downtime, disruption, incompatibility, removed features, increased cost, performance regression, changed access paths, deprecated syntax, and supporting software that is not yet ready.

Key Points: An effective upgrade strategy balances benefits against risks.

Exam Focus: Why have an upgrade strategy? **To balance new-feature benefits with the risks of disruption and incompatibility.**

Factors Affecting the Upgrade Strategy

- **Features and complexity:** How extensive are the changes?
- **Environment complexity:** More servers, users, applications, and locations increase difficulty.
- **Vendor reputation:** A trusted record can increase confidence in a release.
- **Support policies:** Do not leave production on an unsupported release.
- **Organization style:** Type A = aggressive / technology-driven; Type B = middle ground; Type C = conservative / risk-averse.
- **DBA staff skill set:** Experienced staff can make upgrades easier.
- **Platform support:** Different platforms may receive support at different times.
- **Supporting software:** Confirm tools and related applications support the new version.
- **Fallback planning:** Determine whether and how the old version can be restored if problems occur.
- **Migration verification:** Test and verify the system after migration.

Support vs. Exploit

Support: A tool continues to work with the new DBMS release without necessarily using its new features.

Exploit: A tool understands and uses the new features themselves.

Exam Focus: "Supporting a new DBMS release is the same as exploiting it." **False.** Review third-party vendors' support and exploitation policies.

10. Database Standards and Procedures

Standards: Consistent practices intended to improve uniformity and efficiency.

Procedures: Step-by-step instructions for performing a particular task.

Without them, the environment becomes confusing, administration becomes harder, and support costs rise. Greater standardization can reduce user-support costs and improve user satisfaction.

10.1 Database Naming Conventions

Naming conventions promote consistency, easier troubleshooting, easier administration, and smoother migration.

1. Establish conventions for all objects: databases, tables, columns, views, indexes, constraints, triggers, stored procedures, and functions.
2. Minimize name changes across environments. Avoid renaming the same table TEST_EMPLOYEE and PROD_EMPLOYEE. Keep object names stable and distinguish environments through the instance, subsystem, or environment.
3. Use clear, descriptive names, such as HR_EMPLOYEE.
4. Avoid unnecessary restrictions; do not shorten names without a reason.
5. Use standard abbreviations consistently.

Exam Focus: "Standard naming conventions reduce confusion and ease troubleshooting." **True.** "Embedding T for test and P for production in object names is a good idea." **False** under this convention.

10.2 Other Standards and Procedures

Roles and responsibilities: Define who is responsible for each activity.

Communication standards: Define how teams communicate during failures and changes.

Data administration standards: Cover data policies, ownership, stewardship, metadata, and logical models.

Database administration standards: Cover requests to create or modify objects, backup and recovery, and performance tuning.

System administration standards: Cover installation, testing, upgrades, bug fixes, and monitoring.

Database application development standards: Cover SQL coding, SQL performance advice, program preparation, and error handling.

Database security standards: Define who approves privileges, how privileges are granted and revoked, and how governance and compliance are documented.

Exam Focus: "Standards and procedures are only needed for DBAs, not developers or SAs." **False.**

10.3 Application Migration and Turnover Procedures

When multiple environments exist, define how programs, database objects, and structures move between unit testing, integration testing, user acceptance testing, QA, and production.

Multiple environments without clear migration procedures create confusion.

Exam Focus: "When multiple environments exist, clear migration guidelines are required." **True.**

11. DBMS Education

Organizations using DBMS technology should commit to ongoing education for DBAs, programmers, and system administrators.

Course examples include DBMS Overview, Data Modeling and Database Design, Database Administration, Introduction to SQL, Advanced SQL, and Database Programming.

Key Points: Technology changes, so education must continue throughout a career.

Exam Focus: "Organizations using DBMS technology should commit to ongoing technical education." **True.**

Chapter 3 - Data and Storage Management (Book Chapter 18)

1. Storage Management

Storage management concerns the physical media on which DBMS files and data sets reside: disks, SSDs, removable media, optical media, and tape. A DBA must understand physical placement as well as logical database design. Disk is the principal active database medium in this chapter. Database performance depends heavily on I/O; reading from memory or cache is much faster than reading from disk.

Exam Focus: Storage management = physical storage of database data.

2. Main Storage Goals

Preventing data loss is the first priority when selecting storage. Then evaluate capacity, performance, fault tolerance, the ability to add or replace disks without downtime, and cost. The fastest device is not useful if it cannot meet reliability or budget requirements.

3. MTBF

Mean time between failures (MTBF) measures expected time between device failures. A higher MTBF generally indicates better reliability. Mechanical disks can fail, and a larger collection of disks increases the chance that one fails; RAID can reduce the effect of such a failure.

4. Files and Data Sets

A database normally uses several files or data sets: data, indexes, transaction logs, and temporary or work files. A DBA can distribute them across storage devices to improve performance and recovery.

5. File Placement on Disk

File placement chooses the physical device for each file. Separating indexes from data can reduce disk-head contention when a query needs both simultaneously, allowing more efficient I/O.

Exam Focus: The benefit of placing indexes on separate physical disks is reduced disk-head contention and potentially better I/O performance.

6. Transaction Log Placement

The **transaction log** records modifications such as INSERT, UPDATE, and DELETE. Place it on a separate device from database data for performance and recoverability: a failure of the data device should not destroy both the database and its log.

Exam Focus: "The transaction log should be stored on the same device as the database." **False.**

7. Short-Stroking

Short-stroking uses only the outer portion of a disk platter. More data passes under the head per rotation, and the disk arm travels less; the goal is better disk performance.

Exam Focus: "Short-stroking disks can improve disk performance." **True.**

8. System-Managed Storage - SMS

With **system-managed storage (SMS)**, the system determines file locations instead of requiring the DBA to place every file manually. This reduces administrative work. A DBA may still control placement of performance-sensitive files, such as transaction logs.

Exam Focus: "SMS means the system decides file locations instead of the DBA." **True.**

9. Raw Partition

A **raw partition** is a disk device without a file system. The DBMS writes directly to it, avoiding a file-system layer. It was especially useful in some UNIX environments, but raw storage can be harder to track and manage than ordinary files.

Exam Focus: Raw partition = disk device without a file system that permits direct DBMS writes.

10. Raw Partition Trap

Raw partitions may be useful for particular DBMSs and workloads, including logs in some UNIX settings. They are not a universal requirement.

Exam Focus: "DBAs should always place transaction logs only on raw partitions." **False.** The words *always* and *only* make the claim too broad.

11. Temporary Database Files

Temporary files hold short-lived objects and work data used by transactions or queries. They generally do not require the same backup and recovery treatment as permanent data. They may be placed on a separate device and recreated after a failure.

12. Space Management

INSERTs, updates, indexes, logs, and other activity consume space. The DBA monitors free space, fragmentation, partitions, segments, objects nearing their limits, and reserved but unused space. Act before the database runs out of space.

13. Data Page

A **data page** typically contains a **page header** (administrative information), **data rows**, and an **offset table** (pointers to row locations).

Exam Focus: "A database data page includes a page header, data rows, and an offset table." **True.**

14. Can a Row Cross a Page Boundary?

An ordinary row normally remains within one data page. Handling of large values such as text, images, or LOBs depends on the DBMS.

Exam Focus: "A normal row crosses a data-page boundary." **False.**

15. Allocation Page

An **allocation page**, sometimes called a **space-map page**, manages page allocation. It can use bitmaps to identify pages in use, pages with free space, and available pages. An **allocation unit** is a group of pages managed by an allocation page.

Exam Focus: Allocation page = bitmap-based space management; allocation unit = block of pages under that management.

16. Data Record

A **row** is the data, while a **record** includes row data plus housekeeping overhead such as headers and offset information. Storage sizing must account for record length, not merely the sum of column sizes multiplied by row count.

17. Index Page

Indexes have their own pages and records. An index record may contain a header, key values, a pointer to a data row or page, and offset information. Clustered and nonclustered indexes can use different pointer structures. Include index space when estimating total database storage.

18. Transaction Log Sizing

Size the transaction log for peak modification activity and all changes between log archives. A full log can slow or halt processing and endanger recoverability. **Dual active logging** keeps more than one active log copy to reduce the effect of one log failure.

19. Fragmentation

Fragmentation scatters file or index pieces across storage, wasting space and degrading access. Indexes can fragment through inserts, updates, and deletes.

Exam Focus: "Fragmentation affects only the operating system, not database indexes." **False.**

20. SSD - Solid-State Drive

An **SSD** uses solid-state memory instead of mechanical disks, providing faster I/O and avoiding mechanical head movement. In the Second Edition's cost context, SSDs were more expensive and often had less capacity than traditional disks.

21. Tape

Tape is generally unsuitable for active database files. Its primary use is backups and offline storage.

Exam Focus: "Tape is used primarily for backups and offline storage." **True.**

22. RAID

RAID (Redundant Arrays of Independent Disks) combines disks into an array. Different levels emphasize performance, redundancy, fault tolerance, and availability through mirroring, striping, or parity. Some arrays support **hot-swappable drives**, allowing replacement without shutting down the array.

23. Mirroring, Striping, and Parity

Mirroring keeps complete copies on multiple disks for redundancy. **Striping** divides data across disks for performance. **Parity** stores calculated information that can reconstruct data after a disk failure.

24. RAID-0

RAID-0 stripes data for high performance, but provides **no redundancy or fault tolerance**. One disk failure may cause data loss.

Exam Focus: "RAID-0 provides fault tolerance." **False.**

25. RAID-1

RAID-1 mirrors data on two or more disks for redundancy and fault tolerance. Reads can be good; writes require copying data to multiple disks.

Exam Focus: RAID-1 is chosen for critical transaction data because of redundancy and fault tolerance.

26. RAID-2

RAID-2 uses error-correction coding. It has had little significant commercial implementation.

27. RAID-3

RAID-3 uses byte-level striping with a dedicated parity drive. It can provide good data-transfer performance, but implementation is complex.

28. RAID-4

RAID-4 uses block-level striping and one parity disk. That parity disk can become a write bottleneck.

29. RAID-5

RAID-5 distributes parity across drives, avoiding one dedicated parity-disk bottleneck. It can suit read-heavy workloads but is less attractive than RAID-10 or RAID-0+1 for heavy transactional writes.

30. RAID-6

RAID-6 adds a second independent distributed parity calculation for greater fault tolerance. The extra parity work can reduce write performance.

31. RAID-10

RAID-10 combines mirroring and striping for high performance and high fault tolerance, at higher cost. It is a strong choice for heavy transaction activity and mission-critical databases.

32. RAID-0+1

RAID-0+1 combines striping and mirroring to improve speed and reliability. Along with RAID-10, it is a candidate for heavy transaction activity, subject to cost and reliability requirements.

33. RAID Memory Map

RAID-0: striping only; fast, no fault tolerance. **RAID-1:** mirroring. **RAID-2:** error-correction coding. **RAID-3:** byte striping plus dedicated parity. **RAID-4:** block striping plus one parity disk. **RAID-5:** distributed parity. **RAID-6:** dual distributed parity. **RAID-10:** mirror then stripe. **RAID-0+1:** stripe then mirror.

34. Choosing RAID for Different Data

RAID choice depends on workload, reliability, and budget. Read-heavy data may suit RAID-5; write-heavy transaction data may favor RAID-1, RAID-0+1, or RAID-10. Mission-critical high-performance systems often favor RAID-10. A transaction log needs fast sequential writes and reliable protection.

35. JBOD

JBOD (Just a Bunch of Disks) is a simple collection of disks without RAID or SAN features. It is not automatically unsuitable.

Exam Focus: "SAN and NAS are always better than JBOD for every database." **False.**

36. SAN

A **storage area network (SAN)** is a dedicated storage network that can provide high I/O performance, availability, shared storage among hosts, and server/storage consolidation. Its block-level access and low latency suit large, demanding databases.

Exam Focus: SAN = Storage Area Network; fast, reliable shared storage with high I/O performance.

37. NAS

Network-attached storage (NAS) provides **file-level access** over a network, commonly through **NFS** or **CIFS**. It supports file sharing and can simplify management.

Exam Focus: NAS = network file-level storage using protocols such as NFS and CIFS.

38. SAN vs. NAS

SAN: block-level storage infrastructure for high-performance database workloads. **NAS:** file-level network access for shared files and suitable less-demanding storage uses.

39. Tiered Storage

Tiered storage matches data categories with media based on usage, importance, protection, recovery, and performance needs. Its main goal is to reduce total storage cost while meeting service needs. This approach is also called **information life-cycle management (ILM)**.

40. Multi-Temperature Data

Hot data is heavily used and belongs on fast storage. **Warm data** is used less, **cool data** infrequently, and **dormant data** rarely; dormant data may move to tape, optical, or offline media. The general pattern is hot/fast/expensive to dormant/slow/cheap.

41. Capacity Planning

Capacity planning measures current capacity, compares it with requirements, and plans for growth. Track current use, growth rate, new projects, and future storage needs. Capacity also concerns CPU, memory, networks, applications, and system software.

Exam Focus: Capacity planning compares current capacity with requirements and determines when more resources are needed.

42. Four Capacity Questions

The DBA should be able to answer: **When** will more storage be required? **How much** is needed? **Where** is it needed? **What** must be done to align the added storage with the DBMS? Capacity planning is a forward-looking process.

Chapter 4 - Metadata Management (Book Chapter 22)

1. Metadata

Metadata is data about data. It describes a value's type, length, meaning, owner, and use. For example, 56789 is data; "Customer Number is a five-digit numeric field" is metadata. Metadata gives otherwise ambiguous values context.

Exam Focus: Metadata = data about data, including definitions and interfaces.

2. Why Metadata Is Important

Metadata makes data understandable and usable and reduces misinterpretation, misuse, and data-integrity problems. It can answer who, what, when, where, why, and how questions about data.

3. Data to Wisdom

The sequence is **Data > Information > Knowledge > Wisdom**. Data is a fact without context (27). Information adds context (Customer Age = 27). Knowledge combines information with understanding of patterns and relationships. Wisdom applies that knowledge appropriately.

Exam Focus: Wisdom = applied knowledge gained from data and patterns.

4. Metadata Strategy

A **metadata strategy** states how metadata is used, collected, stored, secured, accessed, and measured. It should define usage policy, data ownership and stewardship, required metadata types and purposes, internal and external sources, collection/storage (often a repository), access rules, security, and quality/usability measures.

Exam Focus: A metadata-strategy document outlines policies, ownership, types, collection, and access.

5. Who Leads Metadata Strategy?

Data administration (DA) should lead the metadata effort when that function exists. The **DBA participates** and relies heavily on metadata for technical database work.

6. Data Steward

A **data steward** is accountable for actions involving a defined data set. Stewardship can cover creation, maintenance, and deletion. A steward is not necessarily the data owner.

Exam Focus: "A data steward may create and maintain data but never delete it." **False.**

7. Data Consumer

A **data consumer** directly uses data in daily work. A stewardship policy should identify consumers as well as stewards.

8. Data Warehousing and Metadata

Warehouse users need to know data sources, meaning, transformations, and refresh times. Because warehouse data comes from many sources, high-quality, readily available metadata is essential.

9. Types of Metadata

At the basic classification level used here, metadata has two types: **technology metadata** and **business metadata**.

10. Technology Metadata

Technology metadata explains how data is stored and managed in computer systems: data types, lengths, physical locations, tables, columns, indexes, constraints, files, and security settings. "LICNO is a positive integer from 1 through 9,999,999" describes its technical form.

11. Business Metadata

Business metadata explains what data means and how the organization uses it. "LICNO is a practitioner's license number, must be unique, and each instructor has one" describes business meaning and rules.

12. Technology vs. Business Metadata

Technology metadata: how data is technically stored and managed. **Business metadata:** what data means to the business.

Exam Focus: "Physical storage locations are business metadata." **False.** "LICNO means practitioner license number and must be unique" is **business metadata**.

13. Two Main Types

In this chapter's basic classification, all metadata is either technology or business metadata. The categories are a way to organize meaning, though many organizational roles use both.

14. DBMS System Catalog

The **DBMS system catalog** is a major metadata source for DBAs. It primarily contains technology metadata maintained by the DBMS.

15. What the System Catalog Stores

The catalog records database, table, column, index, and view names; relationships; procedures and triggers; primary and foreign keys; data types and lengths; constraints; physical files; storage and authorization information; and DDL changes.

16. Why the System Catalog Is Useful

Three key catalog characteristics are **active**, **integrated**, and **nonsubvertible**. Together they help keep DBMS object metadata accurate and current.

17. Active

Active means metadata is created and updated automatically when database objects are created or modified. Creating a table, for example, updates its catalog entries.

18. Integrated

Integrated means the catalog is part of the DBMS itself. The DBMS and its object definitions stay connected, supporting accuracy and currency.

19. Nonsubvertible

Nonsubvertible means normal DBMS operations populate the system catalog. Developers do not ordinarily edit catalog rows as regular tables; exceptional direct changes, where allowed, require special care.

Exam Focus: Nonsubvertible = normal DBMS operations are the way to populate the catalog.

20. Is the System Catalog Enough?

No. The catalog is strong on technology metadata but generally lacks complete business descriptions, ownership, non-database files, ETL information, logical models, program usage, and operational context. It is not a complete corporate metadata repository.

21. Metadata Outside the Catalog

Additional metadata may describe flat files; who changed data and when; programs and copybooks that use data; batch jobs and transactions; operations and infrastructure; logical models and physical mappings; ETL sources and transformations; systems of record; and ownership or stewardship.

22. Repository

A **metadata repository** is a centralized store of metadata about an organization's data assets. Its primary purpose is to describe the data, not to store the business data itself.

Exam Focus: "A repository stores metadata about organizational data assets." **True.**

23. Repository Functions

A repository can store data, process, and environment descriptions; support conceptual, logical, and physical views; document assets and produce reports; integrate modeling, ETL, and CASE tools; manage versions and change control; enforce naming conventions; extract metadata from sources or code; and generate copybooks from definitions.

24. Repository and System Catalog Integration

A repository should read the DBMS catalog directly when possible so object descriptions remain current. Otherwise, it needs an effective interface for moving catalog metadata into the repository.

25. Repository Benefits

A central benefit is **consistent documentation of data elements and business rules** across departments. It reduces conflicting names and definitions.

Exam Focus: A common repository benefit is consistent documentation of data elements and business rules.

26. Integrating Multiple Systems

A repository combines metadata from old and new systems so developers can see how data is defined and used across applications.

27. Impact Analysis

Impact analysis shows what a proposed change affects. If a column changes, the repository can reveal dependent programs, reports, ETL jobs, and other assets before implementation.

28. Reusability

Repository documentation helps teams find and reuse existing data or components rather than building duplicates, saving time and resources.

29. Legacy System Integration

Documenting legacy programs, data, and operations helps an organization understand old systems and connect them with newer applications.

30. Data Warehouse Support

Warehouses depend on repository metadata about data sources, transformations, definitions, and usage so users can interpret integrated data correctly.

31. Integrating Data When Populating a Repository

Metadata gathered from multiple systems must be integrated to create a **unified view** and enable cross-system **impact analysis**.

Exam Focus: Integration during repository population supports a unified view across systems and impact analysis.

32. Repository Challenges

The main challenge is **keeping the repository current** as tables, programs, models, ETL processes, and other sources change.

Exam Focus: A repository can become outdated as its source systems change.

33. Populating the Repository

Population gathers metadata from application development tools, code libraries, business documents, modeling tools, the DBMS catalog, ETL and operations tools, and query tools. The process is complex and should be automated as much as possible.

34. Metadata Sources

Sources include application programs and development tools, business users and documents, modeling tools, DBMS catalogs, ETL/warehouse tools, operations and job schedulers, and query-usage tools.

35. Metadata Synchronization

A source may change before the repository's next scan, capture, and integration cycle. During that gap, repository metadata is **out of sync** with the source.

36. Outdated Repository

An outdated, inaccurate, or incomplete repository loses value and can mislead users who rely on it. Currency and accuracy are essential.

37. Repository Requires Commitment

A successful repository requires budget, organizational commitment, procedures, and skilled staff such as data administrators and DBAs. Technology alone is insufficient.

38. Data Dictionary

A **data dictionary** is an older tool focused primarily on data definitions. Earlier dictionaries often required manual entry of definitions.

39. Data Dictionary vs. Repository

A **dictionary** concentrates on definitions; a **repository** handles broader metadata about processes, programs, models, business rules, versions, operations, and relationships.

Exam Focus: "The data dictionary is an older tool that evolved into the repository." **True.**

40. Evolution from Dictionary to Repository

As organizations needed more than definitions, CASE tools and development processes began collecting metadata about which applications and programs use which data. The broader collection evolved into repository technology.

41. DBA's Role in Metadata

The DBA may help select, install, and maintain a repository and contribute to metadata strategy. DBAs often **use** metadata more than they store or administer it.

42. DBA vs. DA in Metadata

DA emphasizes strategy, repositories, business definitions, and policies. **DBA** emphasizes DBMSs, physical databases, catalogs, and technical metadata. Both roles need technology and business metadata.

43. Business Metadata and the DBA

DBAs need business rules to design constraints, keys, relationships, and security correctly. Business users may also need technology metadata. Neither role uses only one metadata type.

Exam Focus: "A DBA has no need for business metadata." **False.**

Chapter 5 - Database Connectivity (Book Chapter 21)

1. Multitier, Distributed Computing

Multitier computing distributes application work across computers or tiers. A client presents the interface, an application server can run business logic, and a database server manages data.

2. Business Issues

Three business forces behind client/server and multitier computing are **downsizing**, **upsizing**, and **rightsizing**.

3. Downsizing

Downsizing moves processing from mainframes toward networked PC workstations. The direction is mainframe to smaller computers.

4. Upsizing

Upsizing connects previously stand-alone PCs so they can reach corporate databases over a network.

5. Rightsizing

Rightsizing places each application on the platform best suited to its requirements. Availability and scale may favor a larger system, while a specialized user interface may favor a PC.

6. Client/Server Computing

Client/server computing is a relationship in which computing entities cooperate on a task. The client requests a service; the server fulfills the request.

7. Client

A **client** is a connected device or program that can present the user interface, send requests, and access network resources.

8. Server

A **server** combines hardware and software to answer requests from multiple clients. It needs processing and networking capacity to handle concurrent work promptly.

9. Four Common Server Types

Common types are **print servers, file servers, database servers, and application servers**.

10. Print Server

A **print server** lets network users share printers rather than requiring a printer for each device.

11. File Server

A **file server** stores and retrieves files. It often transfers a requested file over the network for client-side work rather than processing the client's database query.

12. Database Server

A **database server** accepts a query such as SQL and can return only the requested rows or values. This differs from a file server that may transfer an entire file.

Exam Focus: File server = files; database server = targeted data through queries.

13. Application Server

An **application server** provides services to business applications and centralizes reusable business logic instead of duplicating it on every client.

14. Three Client/Server Software Layers

The layers are **presentation logic**, **application or business logic**, and **data management or DBMS**.

Exam Focus: Name all three layers and the work assigned to each.

15. Presentation Logic

Presentation logic handles what the user sees and interacts with: screens, menus, GUI elements, drag-and-drop, and interactive help.

16. Business Logic

Business logic implements policies, rules, calculations, and transactions that carry out business tasks.

17. Data Management

Data management handles structured data, database access and modification, security, and integrity through the DBMS.

18. Three-Layer Memory Map

Presentation = screen. Business = rules. Data management = DBMS.

19. Centralized Processing

Centralized processing keeps most processing and data on one central computer, traditionally a mainframe accessed through terminals.

Exam Focus: "Centralized processing distributes data across many computers." **False.**

20. Distributed Processing

Distributed processing uses multiple computers to fulfill a request. It places work on suitable platforms and requires a network to connect them.

21. Cooperative Processing

Cooperative processing is distributed processing in which computers share duties. One node might perform basic validation while another applies complex business rules.

22. Types of Client/Server Applications

Client/server models differ in where they place the three layers: presentation, business logic, and data management. Understand each placement rather than memorizing a picture.

23. Distributed User Presentation

In **distributed user presentation**, presentation work is divided between client and server, while business logic and data management remain on the server.

24. Decentralized User Presentation

In **decentralized user presentation**, all presentation logic runs on the client; business logic and data management remain mainly on the server.

25. Distributed Business Logic

Distributed business logic splits rules between client and server. Implementation and maintenance become harder because the code runs on more than one platform.

26. Centralized Data Management Model

Here, the **client** runs presentation and application logic, while the **server** manages data requests and modifications. Centralized DBMS control supports security and integrity.

27. Distributed Data Management

Distributed data management keeps data on both client and server. Local queries may be faster, but administration of multiple DBMS locations is more complex.

28. Multitier Implementation

A common model is **client = presentation**, **application server = business logic**, and **database server = data management**: client > application server > database server.

29. Thin Client

A **thin client** holds limited processing logic; servers handle most work. This simplifies management and upgrades across many clients.

30. Fat Client

A **fat client** holds substantial processing logic. Updating business logic on many client machines can be difficult.

31. Thin vs. Fat Client

Thin = less client logic, easier centralized management. Fat = more client logic, more difficult distributed updates.

Exam Focus: "A thin client contains more processing logic than the server." **False.**

32. Network Traffic

Network traffic is a major cause of poor client/server performance. Moving excessive data or making many round trips increases delay. Send only the needed data.

Exam Focus: A primary client/server performance problem is **network traffic**.

33. Reducing Network Traffic

Return only necessary rows and columns, reduce request count, combine requests when possible, and perform appropriate work near the data.

34. Stored Procedures and Network Traffic

A **stored procedure** can perform several operations on the server after one client call, then return only the necessary result. This reduces network round trips and bandwidth use.

Exam Focus: "Stored procedures can reduce network traffic through server-side processing." **True.**

35. Database Gateway

A **database gateway** lets clients access data on database servers across a network. It is particularly useful when systems use different DBMS products.

36. Gateway in Heterogeneous Databases

A gateway can provide a unified access interface to multiple DBMS products such as Oracle, DB2, and SQL Server.

Exam Focus: Gateway purpose = access different DBMS products through a common interface.

37. Gateway Selection

Select a gateway that supports the required **DBMS products** and **network protocols**, such as DB2 and Oracle over TCP/IP or SNA.

38. Who Configures the Gateway?

The **DBA** usually installs and configures the gateway, often with a **network specialist**. Configuration can be a demanding DBA task.

39. Gateway on Client vs. Server

Installing a gateway on every client makes upgrades harder. A **central gateway server** simplifies updates, though requests must pass through that server.

40. Gateway Features

Important features include **location transparency, commit transparency, two-phase COMMIT, automatic data-type translation, and query/update support**.

41. Location Transparency

Location transparency hides the physical location of data. A user references a table or object without needing to know where it resides.

42. Commit Transparency

Commit transparency lets local and distributed transactions use standard SQL COMMIT without a different user procedure for each location.

43. Two-Phase COMMIT

Two-phase COMMIT coordinates a transaction across multiple databases so participants commit consistently rather than leaving one committed and another failed.

44. Automatic Data Type Translation

A gateway can translate differing DBMS data-type representations so data moves correctly between systems.

45. Database Driver

A **database driver** uses a standard API such as ODBC or JDBC to translate application calls into DBMS-specific requests, send SQL, receive results, and return them to the application.

Exam Focus: "A driver translates API calls into database-specific requests." **True.**

46. Other Driver Functions

Drivers may also handle data conversion, buffering, errors, mappings, load balancing, and connection pooling. Driver choice can strongly affect performance.

47. ODBC

Open Database Connectivity (ODBC) is a standard interface for accessing database servers. An application calls ODBC functions, and an ODBC driver translates them for the DBMS.

48. JDBC

Java Database Connectivity (JDBC) provides database access for Java applications. JDBC drivers are commonly classified as Types 1 through 4.

49. JDBC Type 1

Type 1: JDBC-to-ODBC bridge. It converts JDBC calls through ODBC before reaching the database. It may depend on local ODBC/native libraries, add overhead, and lack newer features.

50. JDBC Type 2

Type 2: native/client-based driver. It uses DBMS vendor client software on each client. Installation, configuration, maintenance, and upgrades must account for those client components.

51. JDBC Type 3

Type 3: network-protocol driver. Calls pass through middleware between the application and DBMS. Vendor DBMS software is unnecessary on the client, but the middle tier contains database-specific code.

52. JDBC Type 4

Type 4: database-wire driver. The JVM connects directly to the DBMS without an ODBC bridge, vendor client software, or special middle tier. It is often favorable for performance and maintenance.

Exam Focus: Type 1 bridges JDBC to ODBC; Type 4 connects directly from the JVM to the DBMS. "Type 4 uses another client-side layer." **False.**

53. Four Driver Memory Map

Type 1 = bridge/ODBC. Type 2 = native client. Type 3 = middleware. Type 4 = direct DBMS wire connection.

54. Connection Pooling

Creating a new database connection for every request is costly. **Connection pooling** reuses an existing connection and returns it to a pool after use instead of closing it permanently.

55. Connection Pool

A **connection pool** is a cache of reusable database connections that reduces connection-creation overhead and improves performance. An application server or driver can manage it.

56. Connection Pool Parameters

Typical settings include initial, minimum, and maximum pool sizes and the time connections remain available.

57. Databases and the Internet

An Internet-accessible database needs a network connection that supports Web access. An organization can host its own database or use an ISP or service provider.

58. Self-Hosting

Hosting the database internally gives more control over data and security. Internal users may retain access even when an external Internet connection fails.

59. ISP Hosting

Provider hosting gives an outside party more control but can lower infrastructure and dedicated-connection costs. It may suit smaller organizations or less critical databases.

60. Internet Database Hosting Guidelines

Use dedicated database-server hardware where practical instead of mixing Web and database workloads on one machine. Separation helps tuning, management, and resource control. Fast network connections support better performance.

61. Internet-Connected Databases

Internet access increases DBA demands, especially **availability**, because users may connect from any location at any time.

62. e-Business

e-Business is online business that users may expect to reach continuously. Operational planning must account for 24/7 use.

63. e-Vailability

e-Vailability is the high availability needed for an always-on Web business: high availability with minimal downtime.

Exam Focus: "E-vailability describes the high availability required by Web-enabled databases." **True.**

64. Enemies of e-Vailability

Downtime and outages prevent continuous service. An eDBA works to minimize both.

65. How to Reduce Downtime

Perform maintenance while systems remain operational, use DBMS availability features, automate DBA work, use fast utilities, apply useful third-party automation, and exploit suitable hardware. Avoid downtime when possible; otherwise shorten it.

66. Automation and Availability

Automation can reduce human errors, maintenance time, and downtime, supporting e-vailability.

67. Internet Time

Web projects often move rapidly. Schedule pressure can lead to poor database design if speed overrides careful planning.

68. Applications vs. Data

Applications change and may be rewritten, but data often survives and moves to successor systems. Do not sacrifice sound database design for a rushed application deadline.

69. Hostile Database

A **hostile database** is hard to understand, query, change, or maintain, often because of poor initial design.

70. Causes of Hostile Databases

Causes include unclear table/column names; designing stored types solely for report appearance (such as storing numbers as characters for leading zeros); packing several facts into one column; volatile primary keys that force foreign-key changes; assumptions that all countries use the same addresses or identifiers; and poor normalization. Use descriptive names, proper domains, stable keys, international awareness, and sound normalization.

71. Web Service

A **Web service** supports machine-to-machine communication over a network so applications can exchange capabilities or data without direct human interaction.

72. SOA

Service-oriented architecture (SOA) separates functions into reusable services accessible over a network.

Exam Focus: Distinct network-accessible functions are the central idea of SOA.

73. SOAP

Simple Object Access Protocol (SOAP) specifies structured messages for Web services. It commonly uses XML and can work over protocols such as HTTP or SMTP.

74. REST

Representational State Transfer (REST) is an architectural style built around resources and representations. A RESTful Web service commonly uses HTTP requests to act on resources and transfer representations.

75. REST vs. SOAP

SOAP is a structured messaging protocol associated with XML. **REST** is a resource-oriented architectural style that commonly uses HTTP and representations.

Exam Focus: "REST uses SOAP as its core messaging protocol." **False.** A RESTful service uses resources and standard Web protocols.

76. Web Services Are Multitier

Web services and REST commonly connect separate application tiers and are examples of multitier processing concepts.

Exam Focus: "Web services and REST are associated with multitier processing." **True.**